



MOXIN LIU

Smart Hardware Project Manager



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[Portfolio: mindscribe-app.pages.dev](https://mindscribe-app.pages.dev)
English working proficiency

PROFILE Smart hardware project manager experienced in the full lifecycle of access control terminals, residential smart locks, glass-door locks, and RV locks—from requirements and R&D tracking through verification, compliance, supplier coordination, pilot builds, mass production, and after-sales closure. Completed an English-taught master’s program and use English as a working language, with independent client coordination experience across Nepal, Pakistan, and other overseas markets. Able to bridge hardware, firmware, mechanical design, cloud/LAN architecture, quality, and supply chain teams to deliver products against milestones and risks.

CORE CAPABILITIES

- Product Lifecycle
- Cross-functional Delivery
- HW/FW Integration
- Verification
- Compliance
- NPI & Mass Production
- Issue Closure
- English Client Communication

PROFESSIONAL EXPERIENCE

Deli Group **Smart Hardware Project Manager** Mar 2026 — Present
Own project execution across access control terminals, residential smart locks, glass-door locks, and RV locks, covering in-house products and OEM programs.

- Managed concurrent 2.4-inch, 5-inch, and 7-inch access control projects and multiple OEM variants; completed development, verification, and pilot build for AL912C, AC213, and AC205C, supporting their August 2026 launch.
- Led type testing, hardware reliability verification, first-round functional validation, and packaging design for the waterproof AC516/AL962C; organized retesting and risk reviews for palm-vein recognition and component lead-time issues.
- Drove new-standard compliance for the AC22 face-recognition lock by comparing certification paths and coordinating corrective actions; also aligned legacy products including DL-AC27 with updated parameters, documentation, and compliance requirements.
- Managed the RV lock from patent-risk screening and supplier sourcing to pricing, pilot verification, and on-site production follow-up; coordinated a supplier fix for low-temperature latch failure, enabling normal sample operation at -30°C.
- Took over a 10-item historical after-sales backlog and closed 5 cases; coordinated root-cause analysis and corrective actions across relay soldering, fingerprint recognition, structural noise, doorbell response, firmware, and power compatibility.
- Produced a focused failure-analysis brief for AC512 doorbell response issues by mapping circuit logic and common wiring errors, improving field diagnosis and after-sales decision consistency.
- Standardized product parameters, operating instructions, technical specifications, and operation-video QR codes; clarified ownership and handoff rules between product and after-sales teams.
- Independently coordinated with customers in Nepal, Pakistan, and other overseas markets, covering requirement clarification, solution alignment, customization follow-up, and project progress communication in English.

DELIVERY METHOD Milestones → risk register → owners → acceptance criteria → verified closure

Guangxi Nuote Broadcasting Co., Ltd. **Thermal Fusion Equipment Project Intern** Jun 2024 — Sep 2024
Coordinated design and factory resources across the product development cycle and helped move three new products from design into pilot production.

Access Terminals 2.4 / 5 / 7-inch and OEM	Residential Locks Product delivery and compliance	Glass-door Locks Recognition, interaction, delivery	RV Locks Mechanism, pilot, production
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DELIVERY CHAIN Definition → R&D alignment → verification → supply chain → production → issue closure



3 models

2.4-inch terminals launched

5 / 10

historical issues closed

-30°C

RV lock validation passed

SELECTED DELIVERY CASES

01 | Multi-model Access Control Launches & OEM Delivery

AL912C / AC213 / AC205C / AC516 / AL962C / AC611 / AC711 / AC612 / AC615

Structured project gates from planning and protocol/functional tests through reliability verification, pilot builds, and warehouse release. Coordinated R&D, quality, sourcing, suppliers, and production across both in-house terminals and OEM firmware/documentation variants.

02 | Smart Lock Compliance Transition

Managed the GB 21556.2-2025 transition for new and in-market products, including AC22 certification and upgrades across AC10, AC21, AC310, and DL-AC27. Balanced launch timing with test non-conformity correction, documentation updates, and legacy portfolio compliance.

03 | Quality & Customer Issue Closure

Established a repeatable flow covering reproduction, impact assessment, hardware/firmware isolation, action ownership, verification, and closure. Applied it to PCB soldering, firmware defects, recognition performance, structural failure, power compatibility, and installation wiring cases.

EDUCATION

Zhejiang Gongshang University Sussex Artificial Intelligence Institute [M.Sc. in Robotics and Autonomous Systems — English-taught](#)

Machine Learning & Deep Learning, Computer Vision, Motion Planning & Control, Embedded AI, 3D Modelling and Mechanical Simulation

Zhejiang Gongshang University [Bachelor's Degree in Electronic Information](#)

Circuit Analysis, Digital/Analogue Electronics, Microcontrollers, Signals & Systems, EDA

ENGINEERING & RESEARCH PROJECTS

- **High-speed railway 5G beam/resource coordination:** Contributed to a four-scenario simulation platform using PPO and Q-learning for continuous beam/power actions and discrete handover decisions; the project report records handover success improving from 92.8% to 97.4% on straight-track simulation.
- **UAV-based cellular coverage blind-spot search:** Built a simulation concept spanning path planning, channel modelling, and signal visualization; inward-spiral paths reduced simulated mission time by approximately 28.92%–33.33% versus reciprocating paths across three scenarios.
- **Candy colour-sorting prototype:** Integrated Arduino UNO, a TCS34725 sensor, and dual servos; the report records 76 pieces sorted in a three-minute accelerated test at 97.4% accuracy, followed by threshold, timing, and anti-sticking improvements.
- **Robotics and autonomous navigation:** Integrated ROS Navigation, Cartographer, and RViz on a DEEP Robotics quadruped platform for live mapping, sensor visualisation, and multi-point navigation; also contributed to five-bar robot and DC motor control projects.

TOOLS, LANGUAGES & DOMAIN KNOWLEDGE

Project Delivery	Milestones, risk registers, cross-functional coordination, supplier management, verification, pilot/mass production, issue closure
Smart Hardware	Access control systems, smart locks, firmware/protocols, LAN/cloud fundamentals, failure analysis, mechanical integration
Tools & Coding Languages	SolidWorks, MATLAB/Simulink, Python, Altium Designer, LabVIEW, C++, TensorFlow/PyTorch
Long-term Habit	Mandarin (native); English (working proficiency) — CET-6 519, CET-4 520, IELTS 6.0 Run approximately 4 km every day, regardless of weather; more than 1,600 km recorded across 2025–2026.

[View product fieldwork, project videos, and the bilingual portfolio](#)

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Target roles: Smart Hardware Project Manager · Product Project Manager · Access Control / IoT / Robotics & Automation